543. Unit digit of $4^1 = 4$

Unit digit of $4^2 = 1\underline{6}$

Unit digit of $4^3 = 6\underline{4}$

Unit digit $4^{96} = 4^{2\times48} = 4^2 = 16$

$\because$ each unit digit of 4 repeats itself after two places

$\therefore$ 16 ÷ 6

$\therefore$ Remainder = 4

544. 67 × 119 + 17 − 27 ÷ 259

= 67 + 119 ÷ 17 × 27 − 259

= 67 + 7 × 27 − 259

= 67 + 189 − 259

= 256 − 259 = −3

545. Given = $\dfrac{(0.73)^3+(0.27)^3}{(0.73)^2+(0.27)^2-0.73\times0.27}$

Let a = 0.73 and b = 0.27

$$=\frac{(0.73+0.27)\left[(0.73)^2+(0.27)^2-0.73\times0.27\right]}{(0.73)^2+(0.27)^2-0.73\times0.27}$$

$$\left[\because a^3+b^3=(a+b)\left(a^2+b^2-ab\right)\right]$$

= 0.73 + 0.27 = 1